

Polymeric Chemistry

DSC and TGA

Universität Stuttgart

Author 1:	Laurens Viehoff
Author 2:	Felix Roth
E-Mail 1:	st183688@stud.uni-stuttgart.de
E-Mail 2:	st188157@stud.uni-stuttgart.de
Student-ID 1:	3676761
Student-ID 2:	3711192
Assistant:	Marcos Dahlem

Version 2

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1 Theory Part

1.1 DSC

The DSC (Differential Scanning Calorimetry) uses the principle of the isotherm calorimetry. The Sample and a references get heated in two separate crucibles. The absorbed energy is measured, and in case of different absorption the one with less gets more power while this added power is being measured and plotted against time (t). The Integral of dW/dt is the difference between the sample and the reference, also shown in picture 1 is the Integral underneath the curve which defines the smelting enthalpy which is calculated by using equation (1). This curve is influenced by the spectrum of lengths the polymer provides, meaning if the polymer has many chains with different lengths the peak gets wider. The reference is always a known Sample or something, for which the heat capacity is known. In our case the reference was the empty crucible.

$$\Delta H_m = \Delta H_{\text{Sample}} - \Delta H_{\text{Reference}} \quad (1)$$

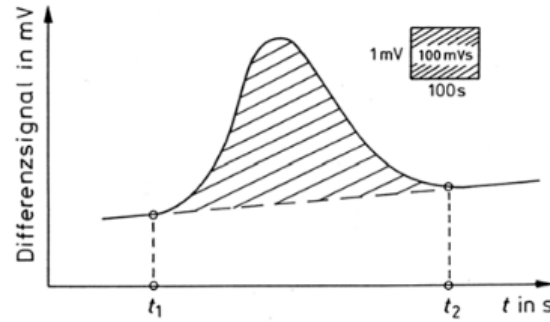


Figure 1: This picture shows the thermogram of a Polymer, the differential signal in [mV] is plotted against time in [s]. [1]

Via this melting enthalpy the crystallization grade can be determined by using the melting enthalpy and by using ΔH_m^0 . ΔH_m^0 cannot directly be determined, because polymers are not fully crystalline. Therefore, an extrapolation is used to determine the ideal case. Meaning the length of the polymer is imagined to be infinite. All of this results in following equation.

$$\alpha = \frac{\Delta H_m}{\Delta H_m^0} \quad (2)$$

1.1.1 Glass transition temperature (T_g) and melting temperature (T_m)

The Glass transition temperature (T_g) is the temperature range where a complete coupled segment movement of chains (10-15 chain links) is possible. These chains were frozen before that temperature and can only be defined if the polymer is amorphous. This can also be seen in a change of the heat capacity.

The melting temperature (T_m) is only present in semi crystalline polymers. These have a lamellar structure (folding crystal) with amorphous parts between the backfolding parts. When reaching

the melting temperature these backfolded parts are moving apart. The melting temperature can be influenced by the amount of amorphous parts in the polymer, meaning if there are more amorphous parts the melting temperature gets higher. It is also influenced by the thickness of the crystall, this is also shown by the following equation:

$$T_m = T_m^0 \cdot \left(1 - \frac{2\sigma}{\rho_c \cdot \Delta H_m^0 \cdot l} \right) \quad (3)$$

In this equation σ is the interfacial tension of the fin surface area, ΔH_m^0 is the specific melting enthalpy, l is the crystall thickness and ρ_c is the density of the crystalline area. T_m^0 is the melting temperature for an ideal crystall and is defined by:

$$T_m^0 = \frac{\Delta H_m^0}{\Delta S_m^0} \quad (4)$$

Both of these temperatures can be analyzed and read out of the thermogram. For semi crystalline polymers with a bigger amorphous share the peak is moved to the right because of the higher temperature.

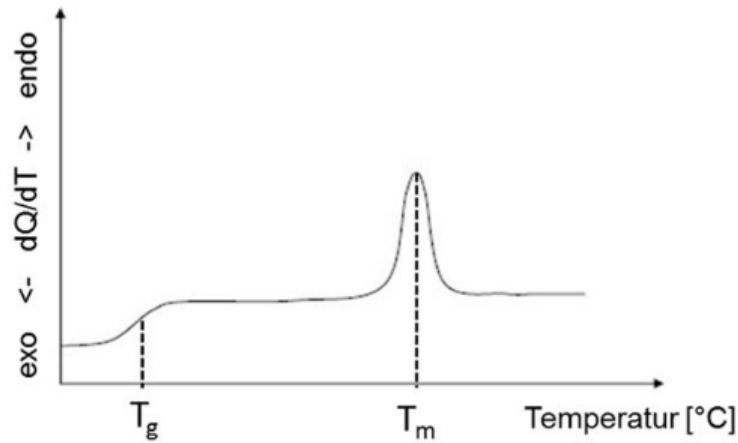


Figure 2: This picture shows an example of a thermogram of a polymer, in which T_g is the Glass transition temperature and T_m is the melting temperature.

1.2 TGA

The main principle of TGA (Thermo gravimetry) is the heating of a Sample to the point where it gets destroyed. In the process the mass is monitored and plotted against temperature. It is also possible to plot the temperature against the time. If these two graphs are compared a clear correlation can be observed.

For analytical purposes the m-T-curve ist the more valuable one, because from this curve things like the starting temperature and the end temperature of the degrading can be determined by tangent constructions. This can also be observed in picture (3).

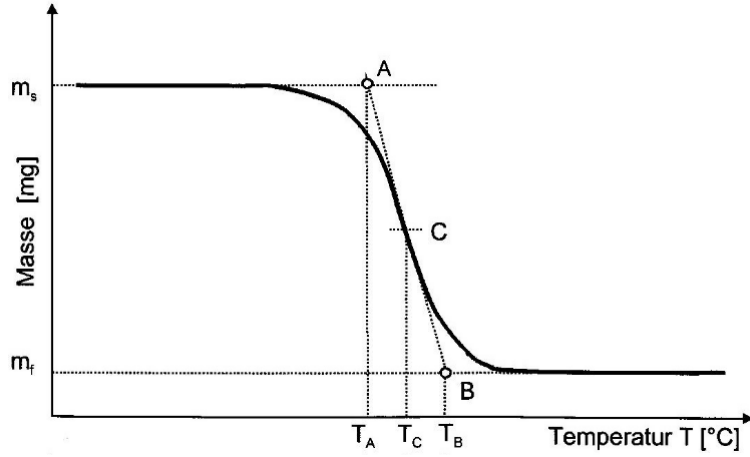


Figure 3: The figure shows the m-T-curve for a polymer. Also shown in the picture are the points A (starting point), B (end point) and point C (middle point) and the three correlating temperatures.

Via this diagram the material specific properties can be characterized.

2 Procedure

To begin with, the samples were placed in small sample vials so that they could be measured. A total of 2.530 mg of PET and 2.328 mg of *it*-PS were used. These samples were then first analysed by DSC; the measurement conditions are listed in Table 1.

Table 1: The measurement conditions for the heating processes for the DSC analyses of PET and *it*-PS samples are listed below.

Sample	Starttemperature [°C]	Endtemperature [°C]	Heating and cooling rate [$\frac{^{\circ}\text{C}}{\text{min}}$]
PET	30	300	20
<i>it</i> -PS	30	300	20

The listed start and end temperatures refer to the heating process, but were used in reverse for the cooling process.

For the TGA measurements, a small, unspecified amount of PET was placed in the apparatus. The sample was then heated using the parameters given in Table 2.

Table 2: The measurement conditions TGA analyses of PET are listed below.

Starttemperature [°C]	Endtemperature [°C]	Heating rate [$\frac{^{\circ}\text{C}}{\text{min}}$]
30	600	20

3 Results

Two different samples were prepared, and analyzed, using DSC and TGA. The Data of the samples (PET and *it*-PS) are listed in the Tables 3 and 4 below.

Table 3: Listed are the melting temperatures (T_m), glass temperatures (T_g) and the melting enthalpy (ΔH_m) from the *it*-PS and PET heating curves.

Sample	$\Delta H_m [\frac{J}{g}]$	$T_m [^{\circ}C]$	$T_g [^{\circ}C]$
PET 1.hc	38.368	249.85	-
PET 2.hc	32.360	246.00	84.36
<i>it</i> -PS 1.hc	24.194	221.61	99.69

Table 4: Listed are the crystallization temperature (T_c) and the crystallization enthalpy (ΔH_c) of the cooling curve of PET.

Sample	$\Delta H_c [\frac{J}{g}]$	$T_c [^{\circ}C]$
PET cc	-34.563	182.73

As shown in the appendix on pages 10 and 11, for *it*-PS were no crystallization enthalpy and no second melting enthalpy observed. Thus, before the measurement both polymers are semi-crystalline, however, the chosen cooling conditions of $20 \frac{^{\circ}C}{min}$ are too fast for the PS to crystallize again and is therefore amorphous during the measurement of the second heating curve. In literature the crystallization of *it*-PS is described as challenging and still studied. They also described the influence of different solvents and cooling rates. As an Exemple they often use cooling conditions of $10 \frac{^{\circ}C}{min}$. [2]

A signal can also be observed in the second heating curve of the *it*-PS, which occurs immediately after the glass transition temperature. This may be due to the fact that, during rapid cooling, the chains are aligned parallel to one another, as shown on the left in Figure 4, but are not crystalline.

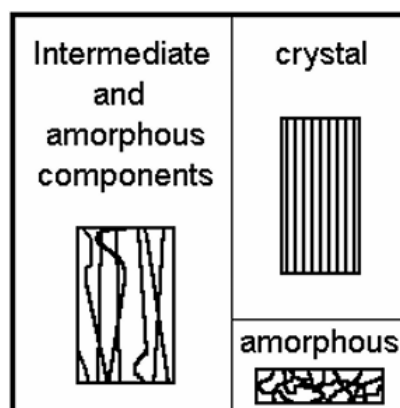


Figure 4: The figure shows possible structural elements of the polymer chains. [3]

Upon reaching the glass transition temperature, the chains move and can align in a partially

crystalline manner. This alignment triggers a melting process, which, occurs at low temperatures. This is because the length of the crystalline regions is very small, and according to equation (3), the melting temperature is very low. [3]

Using the melting enthalpy, the degree of crystallization α were calculated to

$$\alpha_{\text{PET}} = \frac{38,368 \frac{\text{J}}{\text{g}}}{140 \frac{\text{J}}{\text{g}}} = 0.274$$

and

$$\alpha_{\text{it-PS}} = \frac{24,194 \frac{\text{J}}{\text{g}}}{87 \frac{\text{J}}{\text{g}}} = 0.278.$$

For PET the curves can be seen on pages pages 12 to 14. The first curve shows the first heating curve, with an intense and sharp peak at the melting temperature. This is due to the fact that the crystalline parts have roughly the same length. The second curve shows the first cooling curve, with again an sharp and intense crystallization peak. For the last curve, being the second heating curve, we can observe a smaller and broader peak than before. This means that the crystalline parts have a broader spectrum of lengths. Another interesting observation is a little plateau in the melting enthalpy, which is due to an higher amount of same lengths with this melting temperature. An other interesting part is the fact that the glass transition temperature is only visible in the second heating curve, meaning the pure PET sample we measured wasn't in the glass state but somehow been in the elastic state. [1] This can be due to electromagnetic radiation or thermal energy that started the movement of the amorphous phase. With the cooling process in the cuvette the polymer got in the glass state, which is the reason for the observation of the glass transition temperature in the second heating curve. Furthermore, the PET flakes were transparent, which is also a visual indication of amorphous polymers. It therefore stands to reason that the polymer flakes were deliberately produced in an amorphous state so that they could be used for bottles or containers.

The last spektra is the TGA curve of PET, which can be seen on page 15. From 30 to roughly 400 °C the mass stays constant and the polymer stays stable. Afterwards the mass rapidly decreases with increasing temperature, which is due to thermic decomposition. In the end the mass decreased by 86 % and the remaining 14 % were observed as black powder being pure carbon. However there was no plateau visible, which would be the case if there was only carbon left. This can be explained by the destruction of the C-C bonds and the decomposition to smaler hydrocarbon chains. Due to this measurement the PET can be used up to 451,64 °C without significant depolymerization.

4 Questions

4.1 Question 1

The number indicates how many monomers are needed to make one full 360° rotation. The index says how many 360° rotations are needed until the order gets repeated.

For example in the case of 8_5 Polybutene. It needs 8 monomers to complete one 360° rotation and 5 full rotation until the order starts from the start.

4.2 Question 2

it-PP can be made by using the Ziegler-Natta-catalysis. In this case catalysts like TiCl_3 on MgCl_2 -carrier are used in combination with aluminumalkenes. It can also be done via Metallocene-catalysis, this technique is specially used to control the tacticity and results in high graded *it*-PP.

it-PS can also be synthesized using the Metallocene-catalysis. But can also be synthesized via Anionic-Polymerization.

4.3 Question 3

Commonly the degree of crystallization decides how stiff a polymer is. The higher the degree of crystallization, the harder and stiffer the polymer. It also gets less ductile and more dense because crystalline parts can be packed better than amorphous parts.

4.4 Question 4

The Polymer is divided in different sections a back folding section(red), which would be the PE in our case and a amorphous section(black), which in our case would be the PS sections.

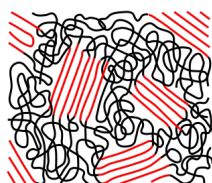


Figure 5: The picture shows the part-.crystalline

4.5 Question 5

To get an semi-crystalline polymer out of an polymerglass it has to be reheated. By reheating the polymer the molecular mobility gets high enough to form crystals. This process is also known as cold crystallization.

Another option is the annealing process. In this process, the polymer is heated slightly above the glass transition temperature, but is not melted. The polymer is then held at that temperature to allow interactions to form. It is subsequently cooled in a controlled manner to preserve the crystalline regions. [4]

4.6 Question 6

The increase in mass can be explained by oxidation. While increasing the temperature in the TGA the polymer oxidizes, the added Oxygen increases the mass of the polymer (molecule if the polymer gets destroyed).

4.7 Question 7

The Mass degeneration curves for different heating procedures are shown in Figure 6.

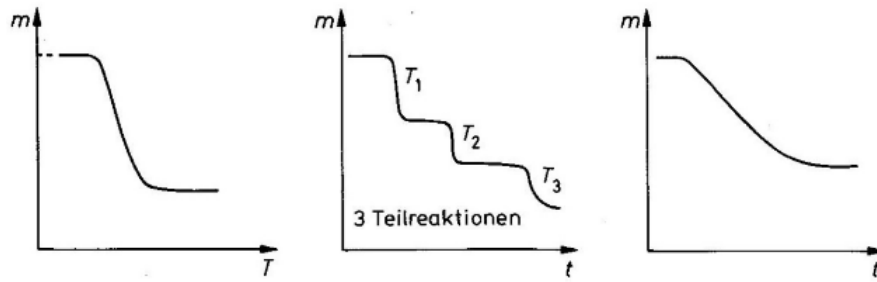


Figure 6: The diagramm on the left visualizes the mass plotted against the temperature. The other two show the mass plotted against time. [1]

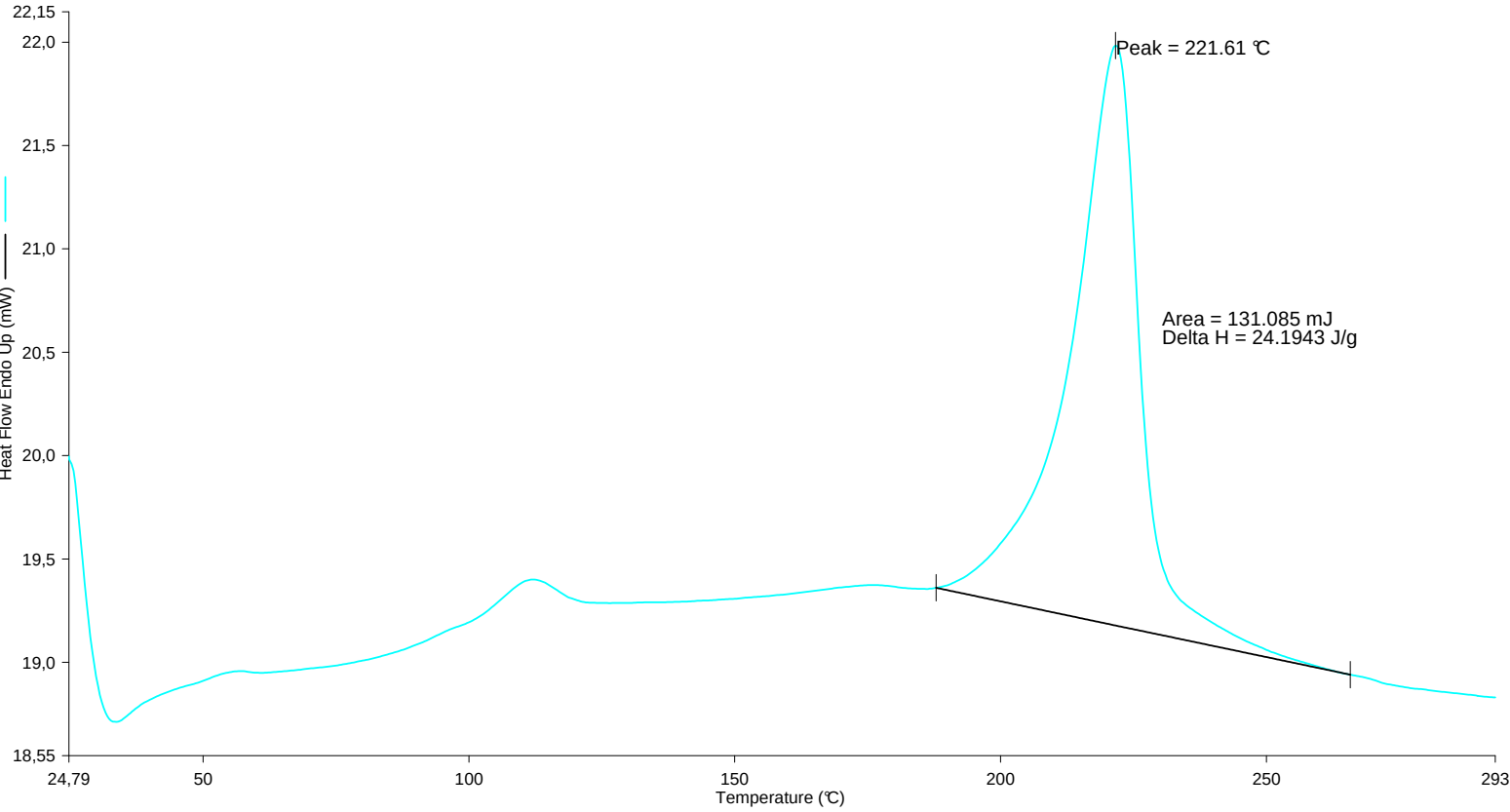
For an constant temperature gradient the graph on the left visualizes the mass degeneration. For an isotherm measurement the right graph visualizes the course and the middle one visualizes an course for an stepwise isothermal procedure. [1]

4.8 Question 8

Due to the residual moisture, the PET decomposes to Ethylenglycol and Terephthalic acid. The Ethylenglycol eliminates water which results in vinylalcohol, which can tautomerize to Ethanal.

5 Appendix

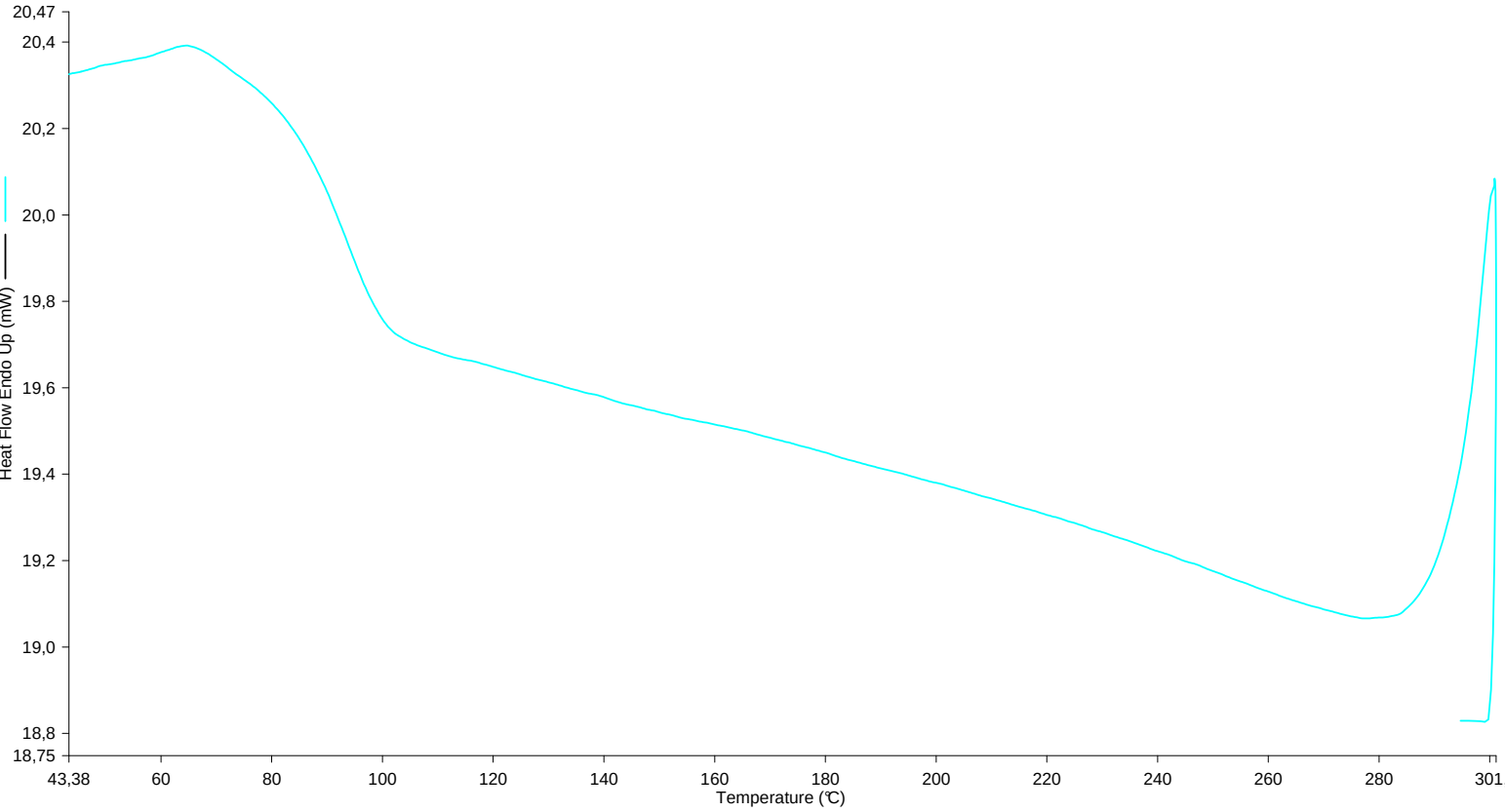
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Operator ID: MD
Sample ID: G8-ps-iso
Sample Weight: 5.418 mg
Comment:



1) Heat from 30.00°C to 300.00°C at 20.00°C/min
2) Cool from 300.00°C to 30.00°C at 20.00°C/min
3) Heat from 30.00°C to 300.00°C at 20.00°C/min

10.03.2026 11:25:16

Filename: C:\Users\DSC\Desktop\MD\G...\ps-iso-dsc.d6d
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Sample ID: G8-ps-iso
Sample Weight: 5.418 mg
Comment:



10.03.2026 11:26:12

- 1) Heat from 30.00°C to 300.00°C at 20.00°C/min
2) Cool from 300.00°C to 30.00°C at 20.00°C/min
3) Heat from 30.00°C to 300.00°C at 20.00°C/min

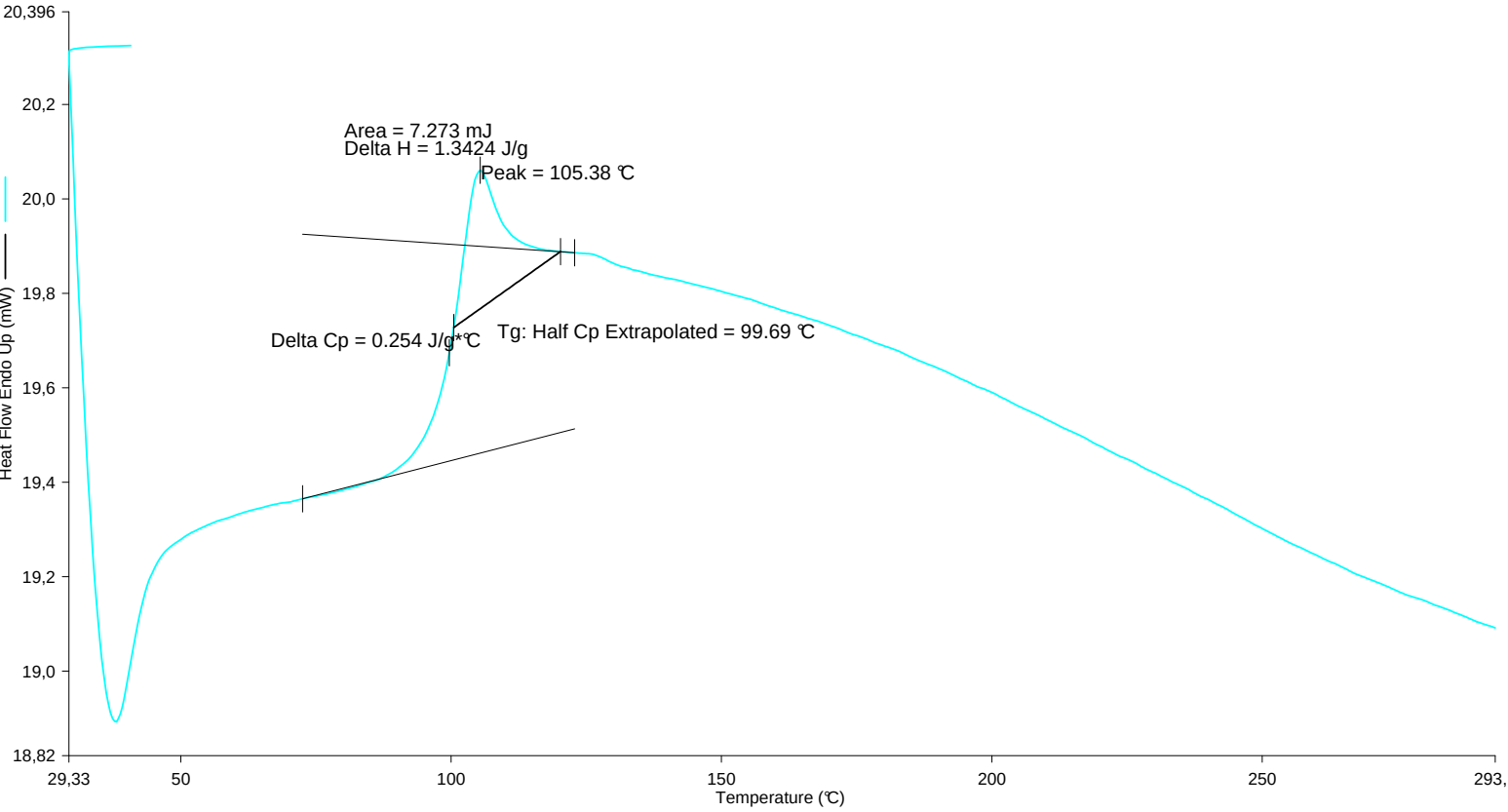
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Operator ID: MD

Sample ID: G8-ps-iso

Sample Weight: 5.418 mg

Comment:



10.03.2026 11:27:53

- 1) Heat from 30.00°C to 300.00°C at 20.00°C/min

2) Cool from 300.00°C to 30.00°C at 20.00°C/min
- 3) Heat from 30.00°C to 300.00°C at 20.00°C/min

Filename:

C:\Users\DSC\Desktop\MD\G8\pet-dsc.d6d

Operator ID:

MD

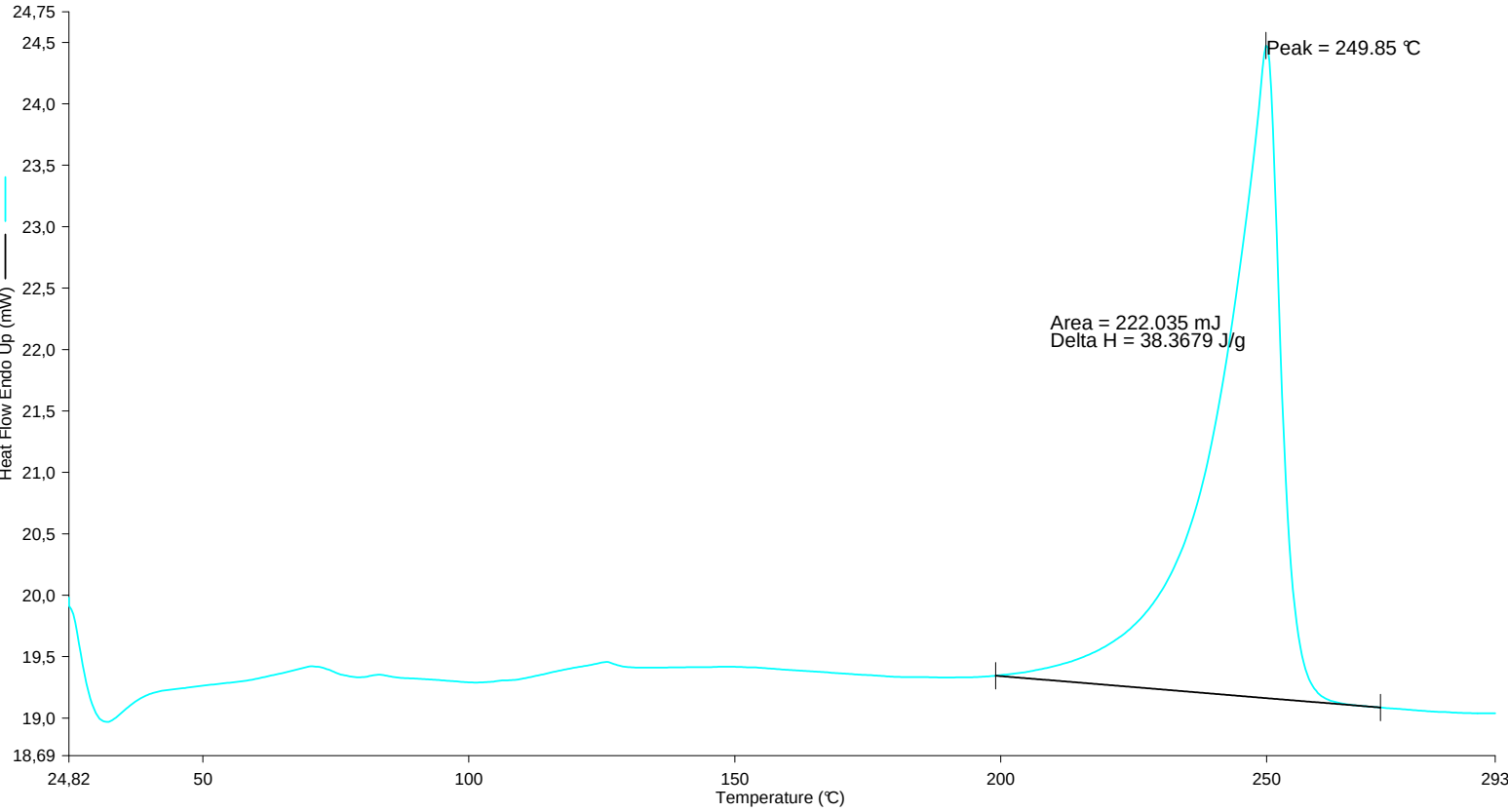
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G8-PET

Sample Weight:

5.787 mg

Comment:



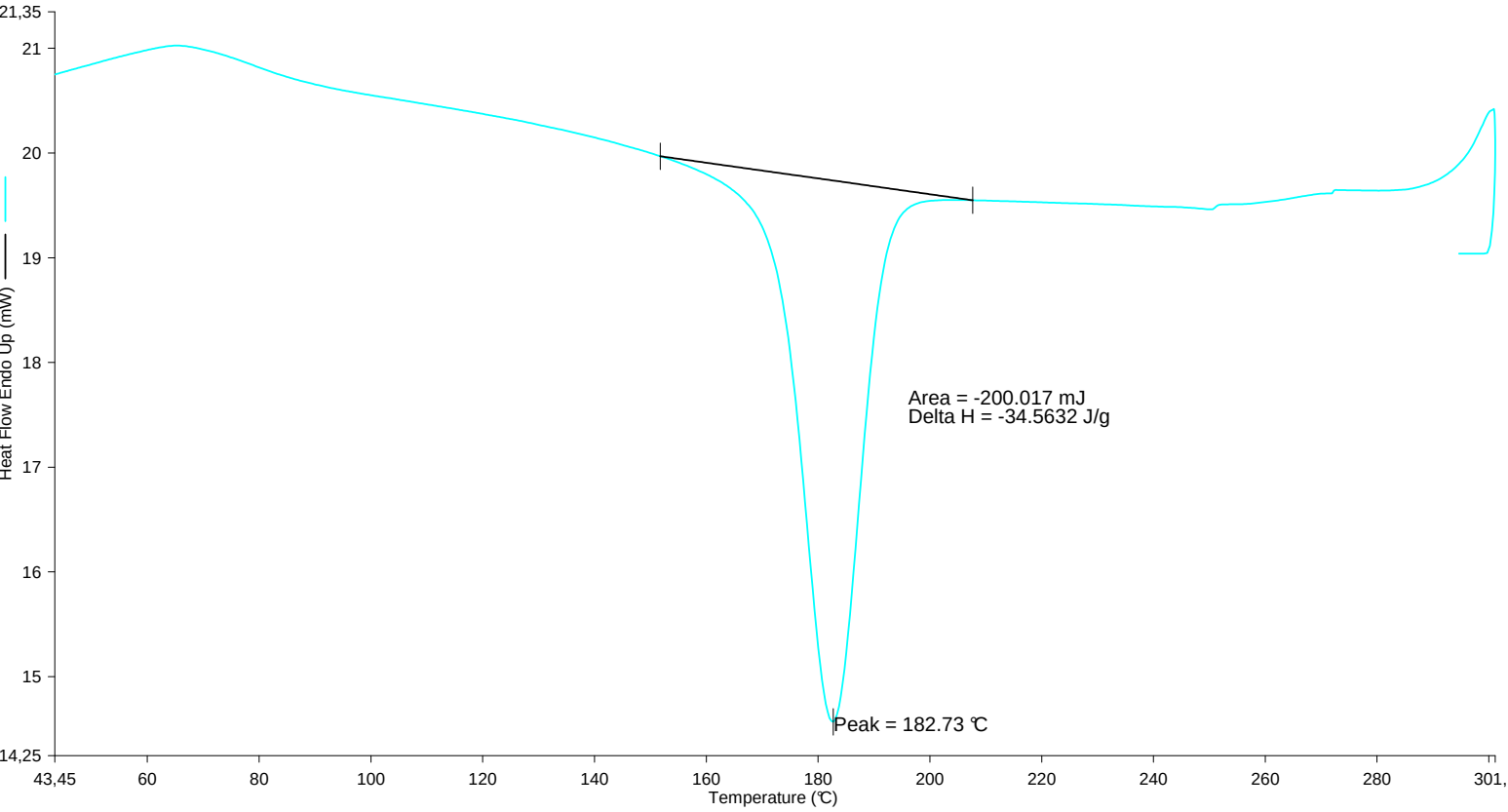
1) Heat from 30.00°C to 300.00°C at 20.00°C/min

2) Cool from 300.00°C to 30.00°C at 20.00°C/min

3) Heat from 30.00°C to 300.00°C at 20.00°C/min

10.03.2026 10:26:23

Filename: C:\Users\DSC\Desktop\MD\G8\pet-dsc.d6d
Operator ID: MD
Sample ID: G8-PET
Sample Weight: 5.787 mg
Comment:



10.03.2026 10:27:39

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2) Cool from 300.00°C to 30.00°C at 20.00°C/min
3) Heat from 30.00°C to 300.00°C at 20.00°C/min

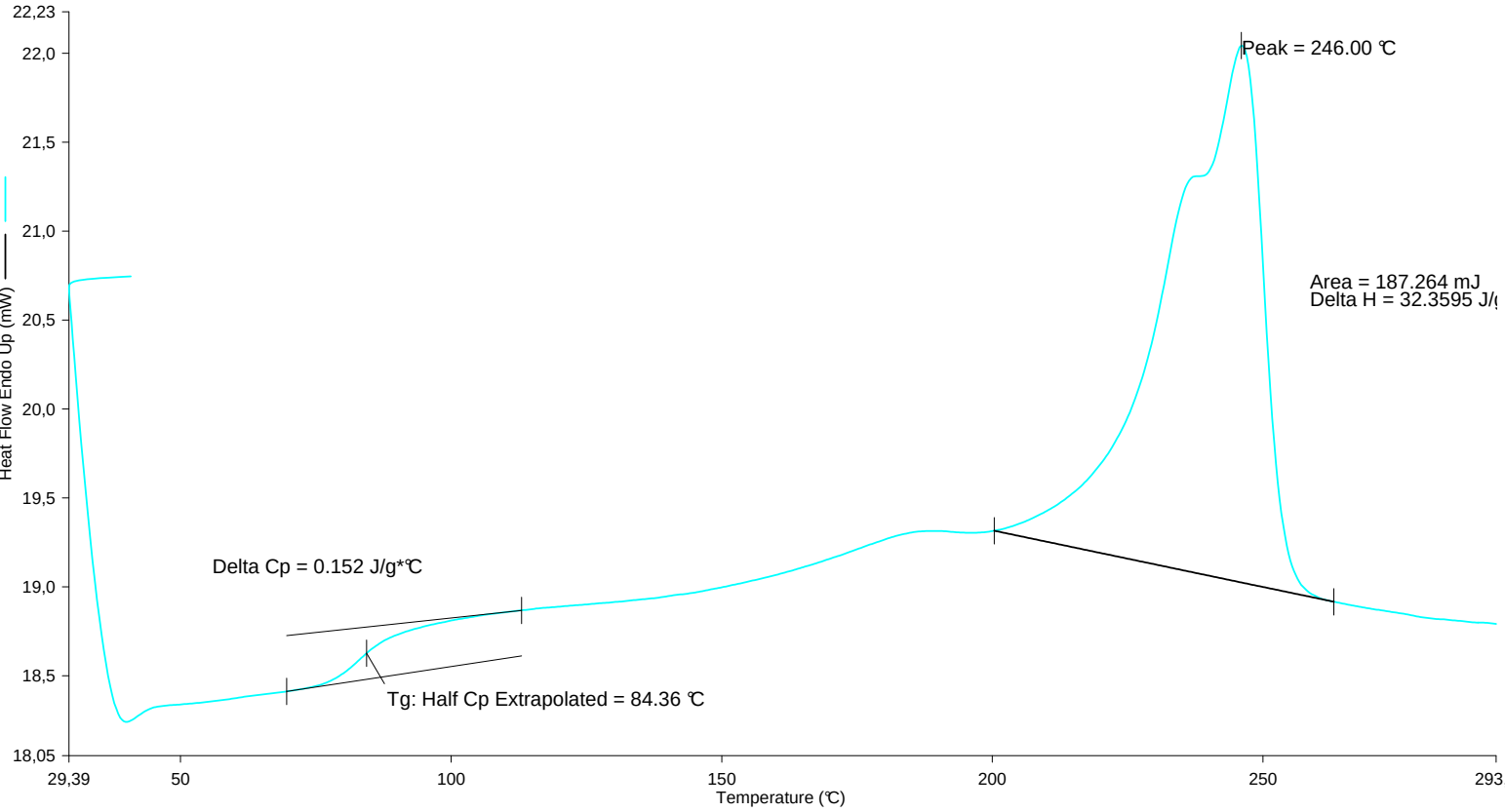
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Operator ID: MD

Sample ID: G8-PET

Sample Weight: 5.787 mg

Comment:



1) Heat from 30.00°C to 300.00°C at 20.00°C/min

2) Cool from 300.00°C to 30.00°C at 20.00°C/min

3) Heat from 30.00°C to 300.00°C at 20.00°C/min

10.03.2026 10:29:27

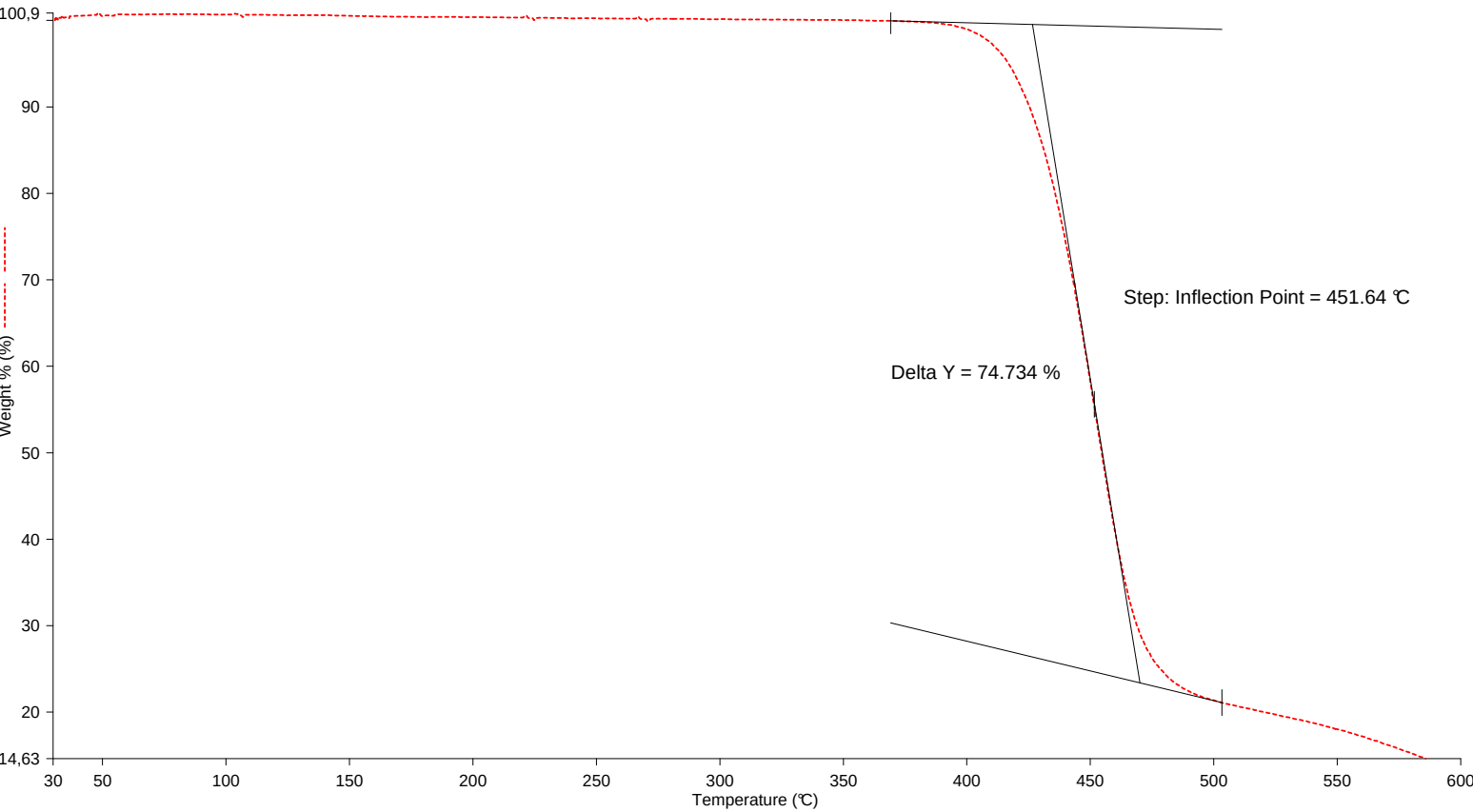
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Operator ID: MD

Sample ID: G8-PET

Sample Weight: 3.597 mg

Comment:



10.03.2026 10:42:27

1) Heat from 30.00°C to 600.00°C at 20.00°C/min

References

- [1] Institut für Polymerchemie - Polymerpraktikum, Vorlesungsskript, Universität Stuttgart, **März 2025**.
- [2] Y. Patil, J. Zhao, M. T. Puthiyaveetil, A. Louhichi, S. Rastogi, *Polymer* **2025**, *326*, 128333.
- [3] In *Thermal Analysis of Polymeric Materials*, (Ed.: B. Wunderlich), Springer, Berlin, Heidelberg, **2005**, pp. 455–590.
- [4] What is Annealing? A Complete Process Guide, en-GB.